Ms07: Kalman Filter II

1D train

$$\mathbf{x}_k = \mathbf{\Phi}_k \mathbf{x}_{k-1} + \mathbf{B}_k \mathbf{u}_k + \mathbf{w}_k$$

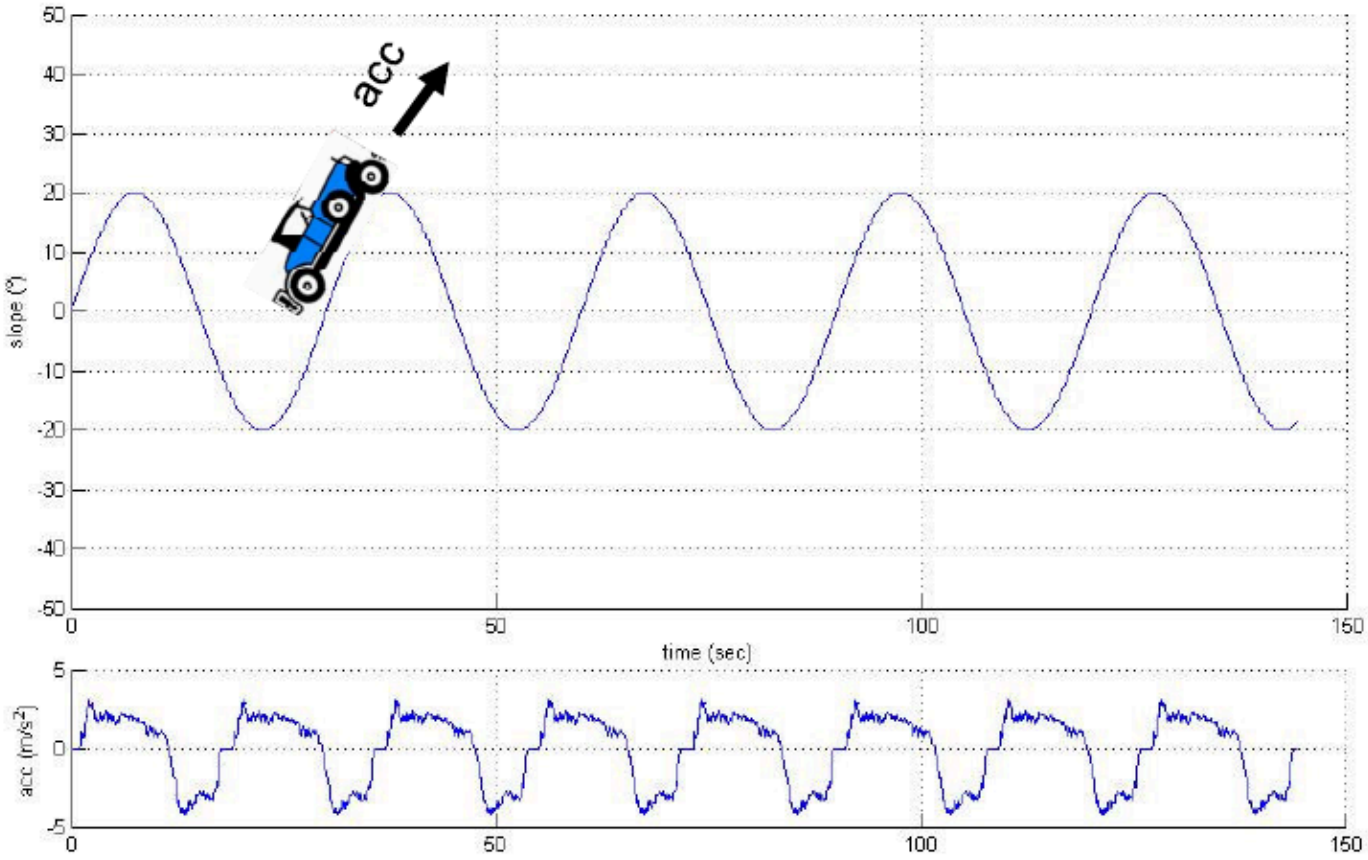
$$\mathbf{x}_k = \begin{bmatrix} r_k \\ \dot{r}_k \end{bmatrix} \qquad \mathbf{u}_k = \frac{f_k}{m}$$

f_k: external applied force (e.g. brake or throttle)

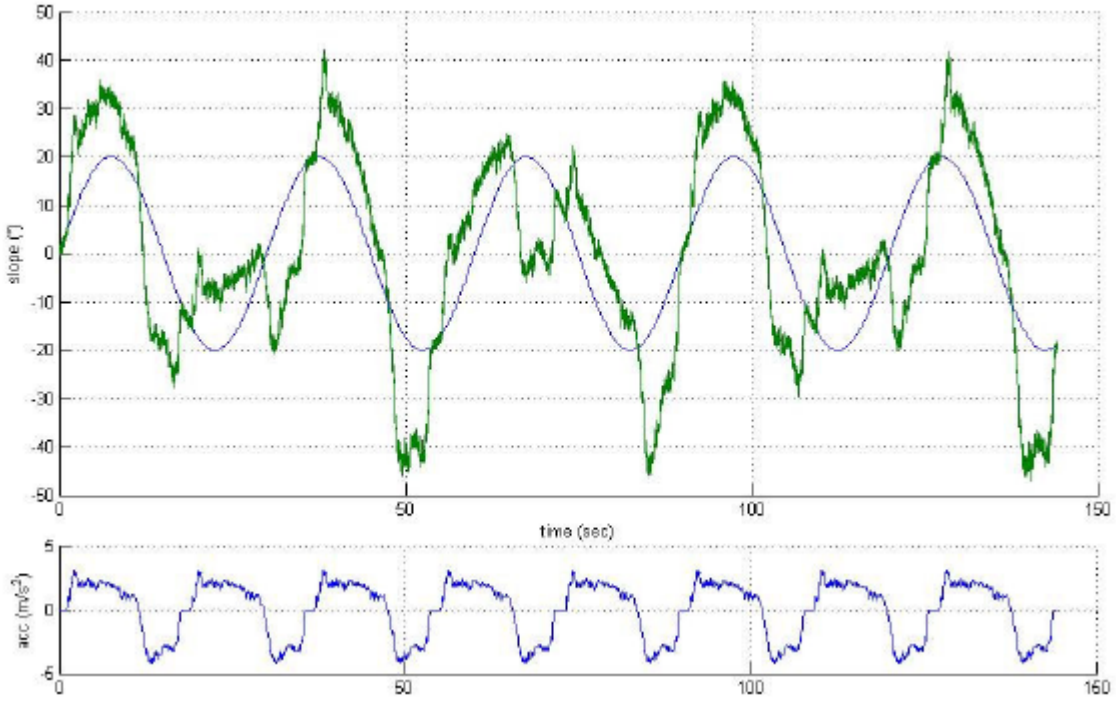
m: mass of the system

$$\underbrace{\begin{bmatrix} r_k \\ \dot{r}_k \end{bmatrix}}_{\mathbf{x}_k} = \underbrace{\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}}_{\mathbf{\Phi}_k} \underbrace{\begin{bmatrix} r_{k-1} \\ \dot{r}_{k-1} \end{bmatrix}}_{\mathbf{x}_{k-1}} + \underbrace{\begin{bmatrix} \frac{1}{2}(\Delta t)^2 \\ \Delta t \end{bmatrix}}_{\mathbf{B}_k} \underbrace{\frac{f_k}{m}}_{\mathbf{u}_k}$$

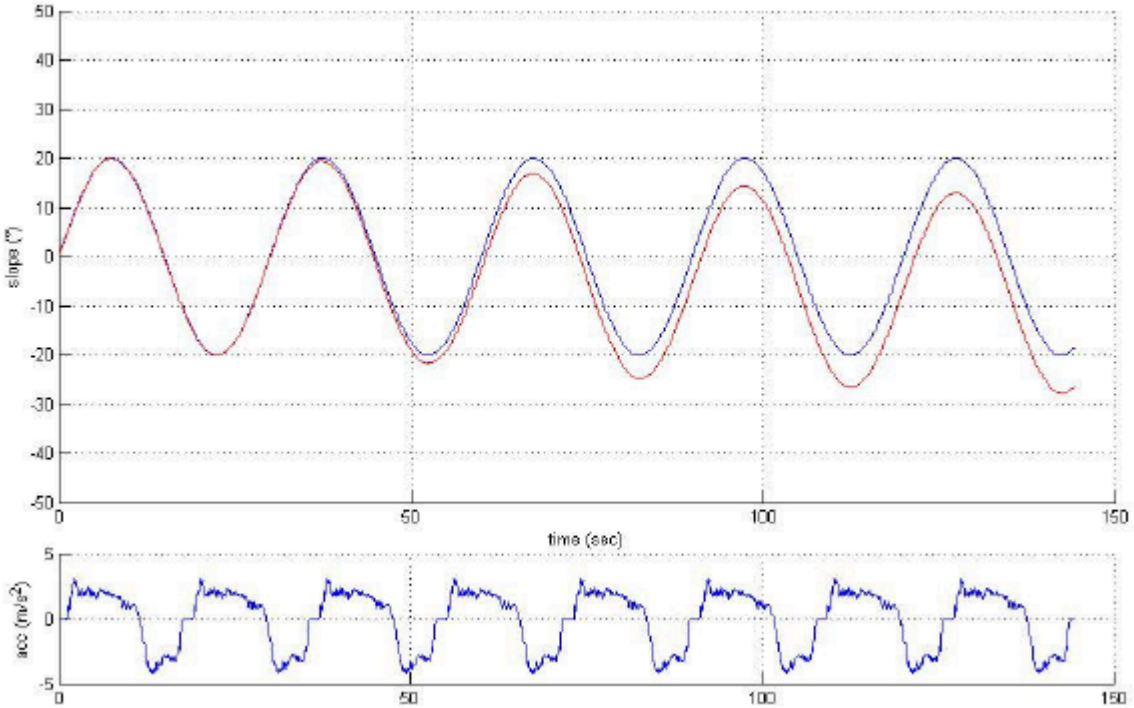
Estimate Inclination of Car

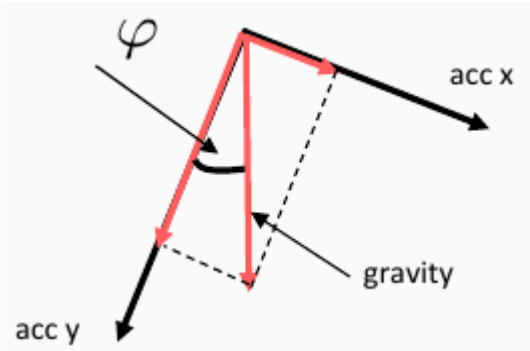


Relative Sensor Readings as Control Input

- **Accelerometer**:
  - $\varphi_{acc} = atan(\frac{acc_x}{acc_y})$
  - **Problem**: Sensitive to additional acceleration due to motion
  - Accelerometer only:
    - Long term stable measurements (drift free)
    - strong short disturbances due to acceleration/deceleration
    - 

- **Gyroscope**:
  - $\varphi_{gyro} = \varphi_0 + \int_{t_0}^{t} w(t)dt$
  - **Problem**:
    - Errors accumulating (drift)
    - Unknown Initial Value
    - Gyroscope only (Strapdown Integration)
      - Accurate short term measurements, but drift over time

- 

- 

  - System model:

$$\begin{array}{ccccccc} \boldsymbol{x}_k & = & \boldsymbol{\Phi}_k & \boldsymbol{x}_{k-1} & + & \boldsymbol{B}_k & \boldsymbol{u}_k + \boldsymbol{w}_k \\ \downarrow & & \downarrow & \downarrow & & \downarrow & \downarrow \\ \underset{\text{angle}}{\varphi_k} & = & 1 & \cdot\varphi_{k-1} & + & \underset{\text{sampling time}}{\tau_s} & \cdot \underset{\text{angular rate}}{w_k} \end{array}$$

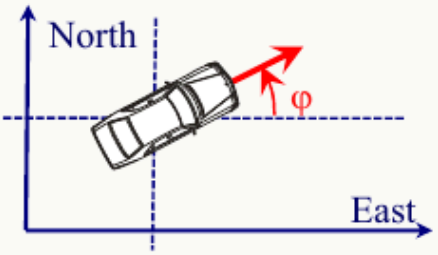
- Measurement Model

$$\begin{array}{c} \boldsymbol{z}_k = \boldsymbol{H}_k \boldsymbol{x}_k + \boldsymbol{v}_k \\ \downarrow \quad \downarrow \\ \varphi_{acc} = 1 \cdot \varphi_k = atan(\frac{acc_x}{acc_y}) \end{array}$$

Kalman Filter to fuse Accelerometer and Gyroscope

- State vector

$$\begin{bmatrix} x \\ y \\ \varphi \end{bmatrix} \quad \begin{matrix} \text{x-position} \\ \text{y-position} \\ \text{angle} \end{matrix}$$



- System model

travelled distance (e.g. from odometer)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \end{bmatrix} = \begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \varphi_{k-1} \end{bmatrix} + \begin{bmatrix} \cos\varphi_{k-1} \cdot \boxed{\Delta t \cdot v_k} \\ \sin\varphi_{k-1} \cdot \Delta t \cdot v_k \\ \boxed{\Delta t \cdot \omega_k} \end{bmatrix}$$

relative angle (e.g. from gyroscope)

- Information about relative motion can be used as control input in the system model
- Can not be written in the form: $\boldsymbol{x}_k = \boldsymbol{\Phi}_k \boldsymbol{x}_{k-1} + \boldsymbol{B}_k \boldsymbol{u}_k$
  - -> **Non-linear system model!**

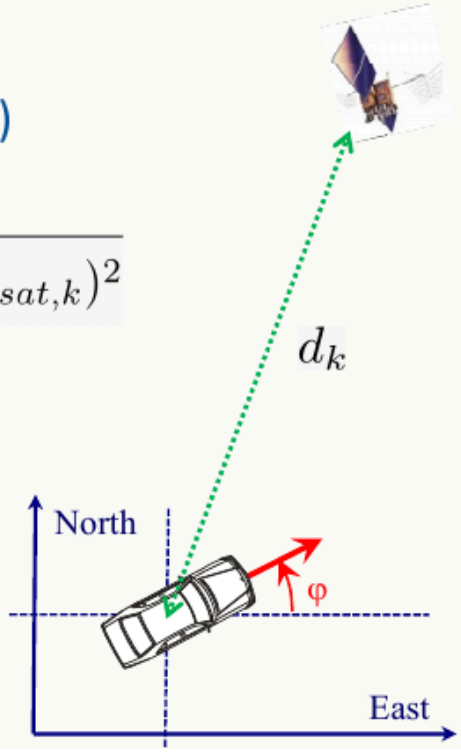
- Measurement Model
  (e.g. GPS, distance to known Position)

$$\mathbf{z}_k = d_k = \sqrt{(x_k - x_{sat,k})^2 + (y_k - y_{sat,k})^2}$$

- Can not be written in the form

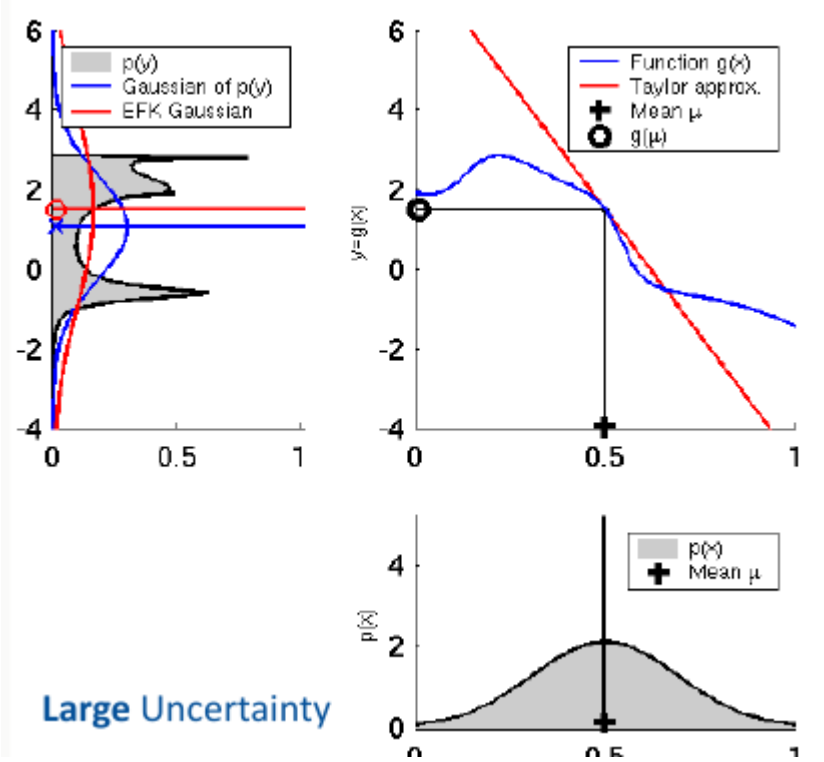
$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k$$

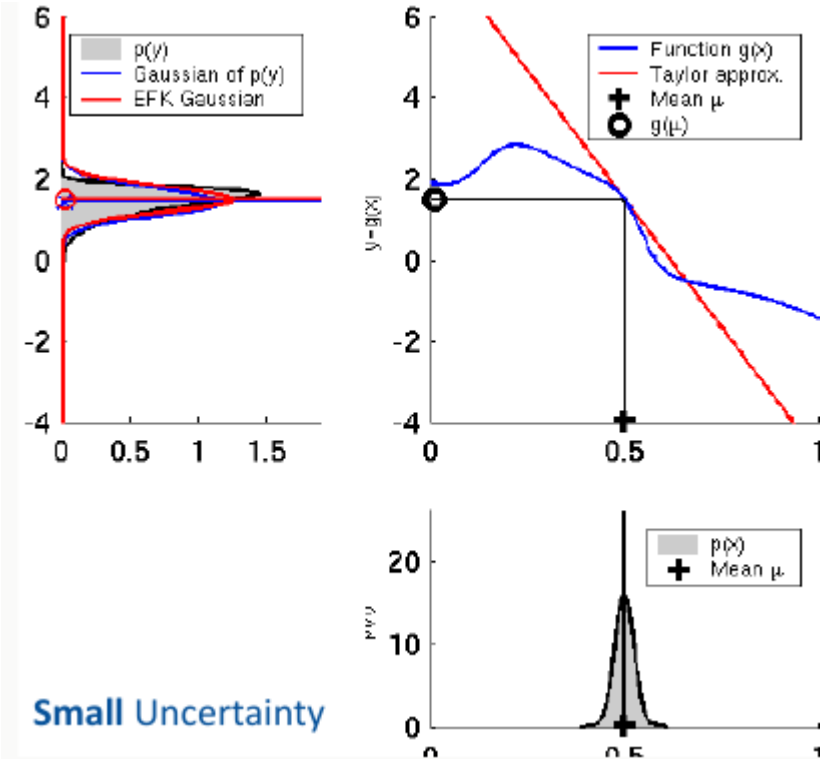
➔ Non linear measurement model



Non Linear Filtering

- So far linear system and measurement models
- General Case:
  - Predicted state is function of last state, control input and noise
    - $\boldsymbol{x}_k = f(\boldsymbol{x}_{k-1}, \boldsymbol{u}_k, \boldsymbol{w}_k)$

- Measurement is a function of the state and noise
  - $\boldsymbol{z}_k = h(\boldsymbol{x}_k, \boldsymbol{v}_k)$
  - 

  - 

- Linearizing the system model

use the previous estimated state as a linearization point

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{w}_k)$$

$$\mathbf{x}_k \approx f(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_k) + \underbrace{\frac{\partial f(\mathbf{x}, \mathbf{u}, \mathbf{w})}{\partial \mathbf{x}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}, \mathbf{u}=\mathbf{u}_k}}_{\Phi_k} (\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1})$$

...

$$\mathbf{P}_k^- = \mathbf{\Phi}_k \mathbf{P}_{k-1} \mathbf{\Phi}_k^T + \mathbf{Q}_k$$

Covariance Update

- Linearizing the measurement model

$$\mathbf{z}_k = h(\mathbf{x}_k, \mathbf{v}_k)$$

$$\mathbf{H}_k = \frac{\partial h(\mathbf{x}, \mathbf{v})}{\partial \mathbf{x}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_k^-}$$

Here we use the predicted state as linearization point

-> **Extended Kalman Filter** using

- non- linear models for
  - state prediction
  - measurement prediction
- Jacobian matrices for
  - Covariance prediction
  - Correction

Prediction

$$\begin{aligned} \mathbf{x}_k^- &= f(\mathbf{x}_{k-1}, \mathbf{u}_k) \\ \mathbf{P}_k^- &= \mathbf{\Phi}_k \mathbf{P}_{k-1} \mathbf{\Phi}_k^T + \mathbf{Q}_k \end{aligned}$$

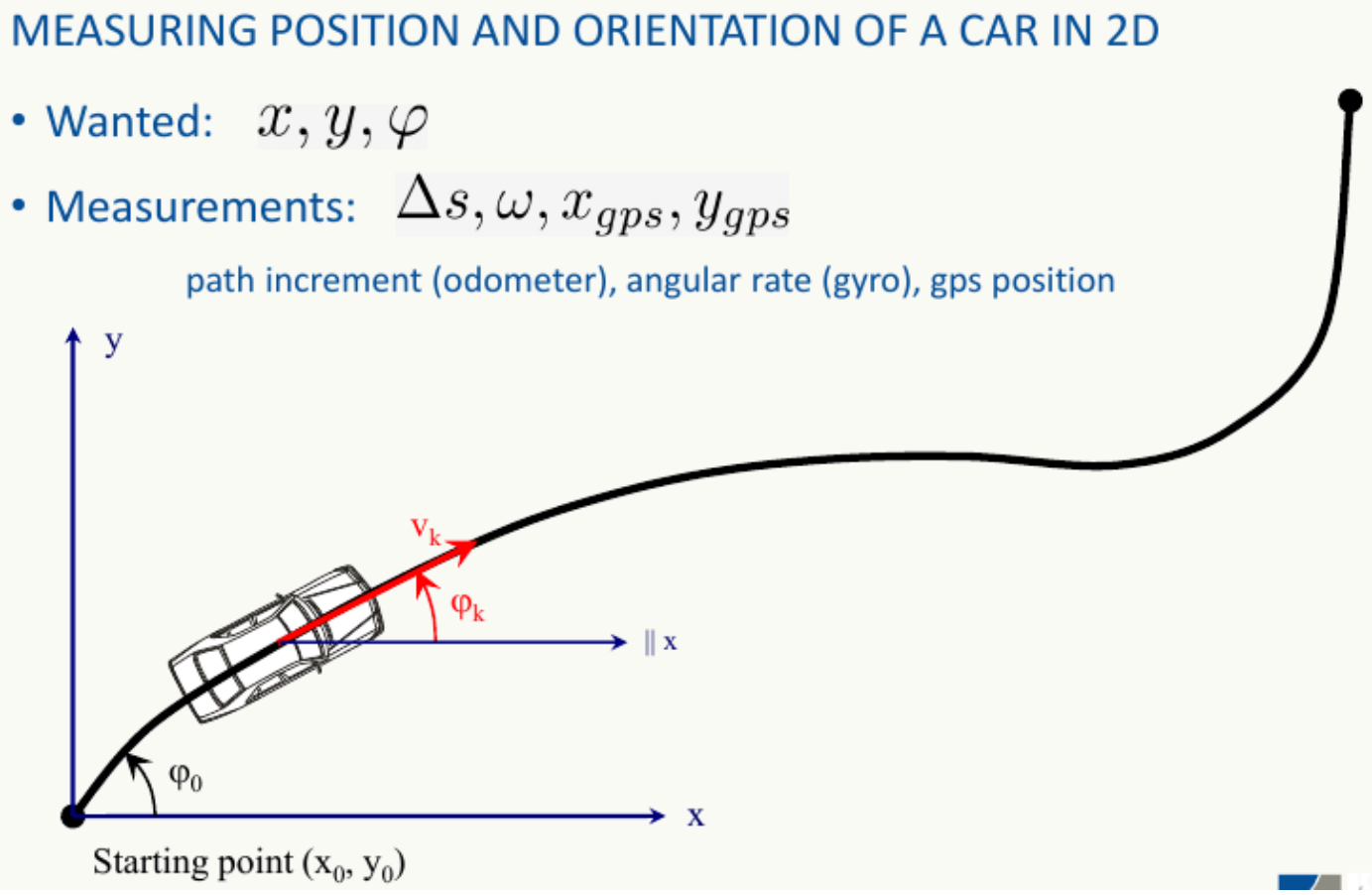
Correction

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}_k^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - h(\mathbf{x}_k^-)) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\mathbf{P}_k^- \end{aligned}$$

$$\mathbf{\Phi}_k = \frac{\partial f(\mathbf{x}, \mathbf{u}, \mathbf{w})}{\partial \mathbf{x}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}, \mathbf{u}=\mathbf{u}_k} \qquad \mathbf{H}_k = \frac{\partial h(\mathbf{x}, \mathbf{v})}{\partial \mathbf{x}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_k^-}$$

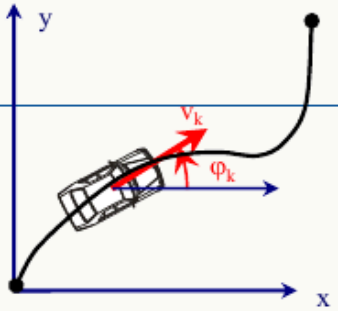
Examples

Example 1



KF EXAMPLE 1

$$\begin{aligned}\mathbf{x}_k^- &= \boxed{f(\mathbf{x}_{k-1}, \mathbf{u}_k)} \\ \mathbf{P}_k^- &= \mathbf{\Phi}\mathbf{P}_{k-1}\mathbf{\Phi}^T + \mathbf{Q}\end{aligned}$$



SYSTEM MODEL

- Assume constant velocity and constant angular rate
- Assume motion is only possible along body frame x-axis
- Linear and angular accelerations are implemented as system noise

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \boxed{\mathbf{w}_k})$$

Non-linear

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \\ \dot{\varphi}_k \\ v_k \end{bmatrix} = f\left(\begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \varphi_{k-1} \\ \dot{\varphi}_{k-1} \\ v_{k-1} \end{bmatrix}\right) = \begin{bmatrix} x_{k-1} + \boxed{\cos(\varphi_{k-1})} v_{k-1}\Delta t \\ y_{k-1} + \boxed{\sin(\varphi_{k-1})} v_{k-1}\Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1}\Delta t \\ \dot{\varphi}_{k-1} \boxed{+ w_{\dot{\varphi}}\Delta t} \\ v_{k-1} \boxed{+ w_v \Delta t} \end{bmatrix}$$

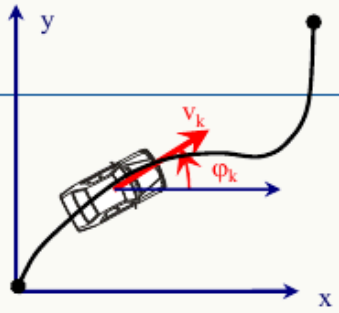
system noise

KF EXAMPLE 1

$$\begin{aligned}\mathbf{x}_k^- &= f(\mathbf{x}_{k-1}, \mathbf{u}_k) \\ \mathbf{P}_k^- &= \boxed{\mathbf{\Phi}\mathbf{P}_{k-1}\mathbf{\Phi}^T} + \mathbf{Q}\end{aligned}$$

TRANSITION MATRIX

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



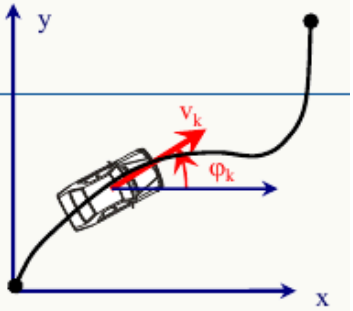
$$\mathbf{\Phi}_{k-1} = \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}} = \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \frac{\partial f_1(\mathbf{x})}{\partial x_2} & \cdots \\ \frac{\partial f_2(\mathbf{x})}{\partial x_1} & \frac{\partial f_2(\mathbf{x})}{\partial x_2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}}$$

$$= \frac{\partial}{\partial \mathbf{x}} \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1}\Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1}\Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1}\Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}}\Delta t \\ v_{k-1} + w_v \Delta t \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}}$$

KF EXAMPLE 1

TRANSITION MATRIX

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



$$\mathbf{\Phi}_{k-1} = \frac{\partial}{\partial \mathbf{x}} \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}} \Delta t \\ v_{k-1} + w_v \Delta t \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}}$$

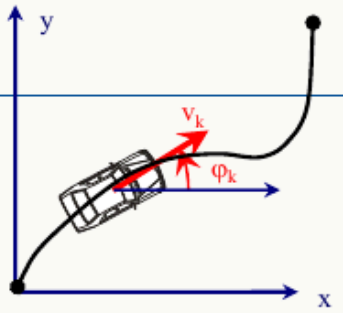
$$= \begin{bmatrix} 1 & 0 & -\sin(\varphi_{k-1}) v_{k-1} \Delta_t & 0 & \cos(\varphi_{k-1}) \Delta t \\ 0 & 1 & \cos(\varphi_{k-1}) v_{k-1} \Delta_t & 0 & \sin(\varphi_{k-1}) \Delta t \\ 0 & 0 & 1 & \Delta t & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}}$$

$$\begin{aligned} \mathbf{x}_k^- &= f(\mathbf{x}_{k-1}, \mathbf{u}_k) \\ \mathbf{P}_k^- &= \mathbf{\Phi} \mathbf{P}_{k-1} \mathbf{\Phi}^T + \boxed{\mathbf{Q}} \end{aligned}$$

KF EXAMPLE 1

SYSTEM NOISE

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



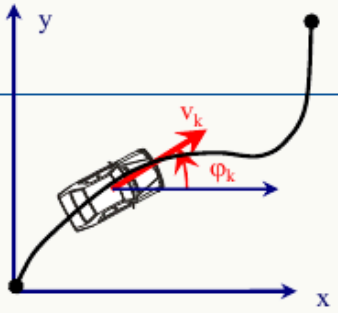
$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{\sigma_{\dot{\varphi}}^2 \Delta t^2} & 0 \\ 0 & 0 & 0 & 0 & \boxed{\sigma_v^2 \Delta t^2} \end{bmatrix}$$

angular acceleration variance

acceleration variance

KF EXAMPLE 1

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



MEASUREMENT MODEL GPS

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

- GPS Position x_{gps}, y_{gps}

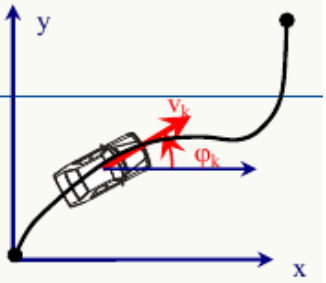
$$\mathbf{z}_{gps,k} = \begin{bmatrix} x_{gps,k} \\ y_{gps,k} \end{bmatrix} = h_{gps}(\mathbf{x}_k) = \mathbf{H}_{gps}\mathbf{x}_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R}_{gps} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 \\ 0 & \sigma_{gps,y}^2 \end{bmatrix}$$

KF EXAMPLE 1

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



MEASUREMENT MODEL PATH INCREMENT

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

- Path increment (from odometer) Δs

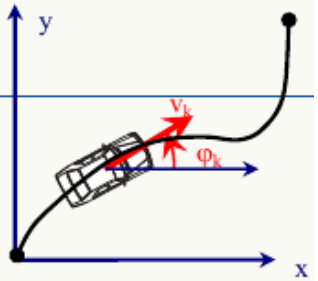
$$\mathbf{z}_{odo,k} = \Delta s_k = h_{odo}(\mathbf{x}_k) = \mathbf{H}_{odo}\mathbf{x}_k = \begin{bmatrix} 0 & 0 & 0 & 0 & \Delta t \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R}_{odo} = \sigma_{odo}^2$$

KF EXAMPLE 1

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



MEASUREMENT MODEL ANGULAR RATE

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

- Angular rate (from gyroscope) ω

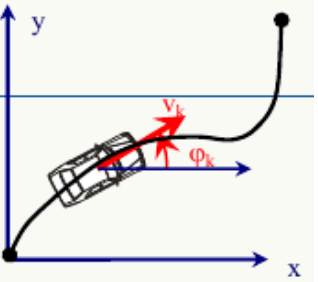
$$\mathbf{z}_{gyro,k} = \omega_k = h_{gyro}(\mathbf{x}_k) = \mathbf{H}_{gyro}\mathbf{x}_k = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R}_{gyro} = \sigma_{gyro}^2$$

KF EXAMPLE 1

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \end{bmatrix}$$



MEASUREMENT MODEL COMBINED

$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

only useful for identical data rates

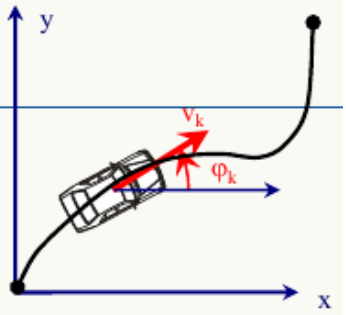
$$\mathbf{z}_k = \begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ \Delta s_k \\ \omega_k \end{bmatrix} = \mathbf{H}\mathbf{x}_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 & 0 & \\ 0 & \sigma_{gps,y}^2 & 0 & 0 \\ 0 & 0 & \sigma_{odo}^2 & 0 \\ 0 & 0 & 0 & \sigma_{gyro}^2 \end{bmatrix}$$

Example 2

KF EXAMPLE 2



SYSTEM MODEL

- Assume constant acceleration and constant angular rate
- Assume motion is only possible along body frame x-axis
- Angular accelerations and linear jerk are implemented as system noise

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \boxed{\mathbf{w_k}})$$

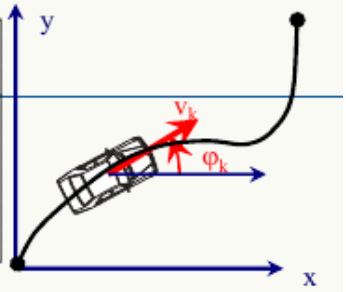
$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \\ \dot{\varphi}_k \\ v_k \\ a_k \end{bmatrix} = f\left(\begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \varphi_{k-1} \\ \dot{\varphi}_{k-1} \\ v_{k-1} \\ a_{k-1} \end{bmatrix}\right) = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + \boxed{w_{\dot{\varphi}} \Delta t} \\ v_{k-1} + a_{k-1} \Delta t \\ a_{k-1} + \boxed{w_a \Delta t} \end{bmatrix}$$

system noise

KF EXAMPLE 2

$$\mathbf{x}_k^- = f(\mathbf{x}_{k-1}, \mathbf{u}_k)$$
$$\mathbf{P}_k^- = \boxed{\mathbf{\Phi} \mathbf{P}_{k-1} \mathbf{\Phi}^T} + \mathbf{Q}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \\ a \end{bmatrix}$$



TRANSITION MATRIX

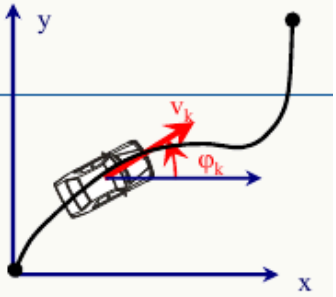
$$\mathbf{\Phi}_{k-1} = \frac{\partial}{\partial \mathbf{x}} \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}} \Delta t \\ v_{k-1} + a_{k-1} \Delta t \\ a_{k-1} + w_a \Delta t \end{bmatrix}_{\mathbf{x} = \hat{\mathbf{x}}_{k-1}}$$

$$= \begin{bmatrix} 1 & 0 & -\sin(\varphi_{k-1}) v_{k-1} \Delta t & 0 & \cos(\varphi_{k-1}) \Delta t & 0 \\ 0 & 1 & \cos(\varphi_{k-1}) v_{k-1} \Delta t & 0 & \sin(\varphi_{k-1}) \Delta t & 0 \\ 0 & 0 & 1 & \Delta t & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 0 & & 1 \end{bmatrix}_{\mathbf{x} = \hat{\mathbf{x}}_{k-1}}$$

KF EXAMPLE 2

$$\mathbf{x}_k^- = f(\mathbf{x}_{k-1}, \mathbf{u}_k)$$
$$\mathbf{P}_k^- = \mathbf{\Phi}\mathbf{P}_{k-1}\mathbf{\Phi}^T + \boxed{\mathbf{Q}}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \\ a \end{bmatrix}$$



SYSTEM NOISE

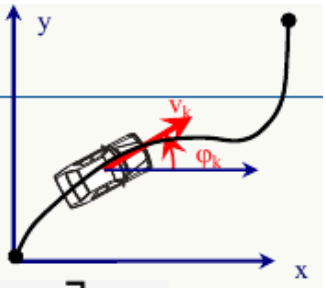
$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \boxed{\sigma_{\dot{\varphi}}^2 \Delta t^2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \boxed{\sigma_a^2 \Delta t^2} \end{bmatrix}$$

angular acceleration variance

jerk variance

KF EXAMPLE 2

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \\ \dot{\varphi} \\ v \\ a \end{bmatrix}$$



MEASUREMENT MODEL

$$\mathbf{z}_k = \begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ a_k \\ \omega_k \end{bmatrix} = \mathbf{H}\mathbf{x}_k = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \mathbf{x}_k$$

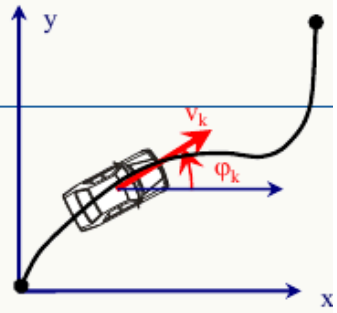
e.g. acceleration from IMU
(rotated into driving direction and corrected for gravity,
see velocity update from strapdown integration)

- Measurement Noise

$$\mathbf{R}_{gps} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 & 0 & \\ 0 & \sigma_{gps,y}^2 & 0 & 0 \\ 0 & 0 & \sigma_{acc}^2 & 0 \\ 0 & 0 & 0 & \sigma_{gyro}^2 \end{bmatrix}$$

KF Example 3

KF EXAMPLE 3



SYSTEM MODEL

- No assumed (and wrong) model on the temporal evolution of motion states
- Relative measurements (path increment and angle change) are modelled as control input
- System noise is represented as control noise

control input

$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \boxed{\mathbf{u}_k}, \boxed{\mathbf{w}_k})$$

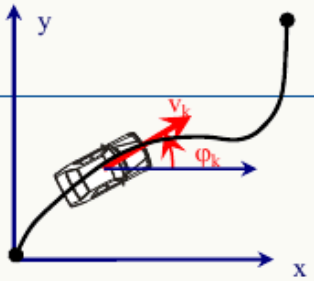
$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \end{bmatrix} = f\left(\begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \varphi_{k-1} \end{bmatrix}, \begin{bmatrix} \Delta s_k \\ \omega_k \end{bmatrix}, \begin{bmatrix} w_{odo} \\ w_{gyro} \end{bmatrix}\right) = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1})(\boxed{\Delta s_k} \boxed{+ w_{odo}}) \\ y_{k-1} + \sin(\varphi_{k-1})(\boxed{\Delta s_k} \boxed{+ w_{odo}}) \\ \varphi_{k-1} + (\boxed{\omega_{gyro,k}} + \boxed{w_{gyro}})\Delta t \end{bmatrix}$$

System/control noise

KF EXAMPLE 3

TRANSITION MATRIX

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \end{bmatrix}$$



$$\mathbf{\Phi}_{k-1} = \frac{\partial}{\partial \mathbf{x}} \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1})(\Delta s_k + w_{odo}) \\ y_{k-1} + \sin(\varphi_{k-1})(\Delta s_k + w_{odo}) \\ \varphi_{k-1} + (\omega_{gyro,k} + w_{gyro})\Delta t \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}, \mathbf{u}=\mathbf{u}_k}$$

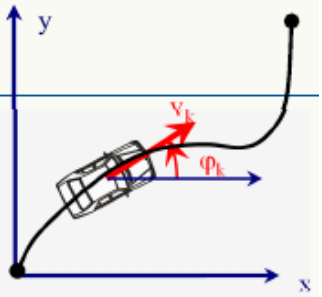
$$= \begin{bmatrix} 1 & 0 & -\sin(\varphi_{k-1})\Delta s_k \\ 0 & 1 & \cos(\varphi_{k-1})\Delta s_k \\ 0 & 0 & 1 \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}, \mathbf{u}=\mathbf{u}_k}$$

KF EXAMPLE 3

$$\begin{aligned}\mathbf{x}_k^- &= f(\mathbf{x}_{k-1}, \mathbf{u}_k)\\ \mathbf{P}_k^- &= \mathbf{\Phi}\mathbf{P}_{k-1}\mathbf{\Phi}^T + \boxed{\mathbf{Q}}\end{aligned}$$

SYSTEM NOISE

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \end{bmatrix}$$



$$\mathbf{x}_k = f(\mathbf{x}_{k-1}, \mathbf{u_k}, \boxed{\mathbf{w}_k})$$

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \end{bmatrix} = f\left(\begin{bmatrix} x_{k-1} \\ y_{k-1} \\ \varphi_{k-1} \end{bmatrix}, \begin{bmatrix} \Delta s_k \\ \omega_k \end{bmatrix}, \boxed{\begin{bmatrix} w_{odo} \\ w_{gyro} \end{bmatrix}}\right) = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1})(\Delta s_k + \boxed{w_{odo}}) \\ y_{k-1} + \sin(\varphi_{k-1})(\Delta s_k + \boxed{w_{odo}}) \\ \varphi_{k-1} + (\omega_{gyro,k} + \boxed{w_{gyro}})\Delta t \end{bmatrix}$$

control noise:

$$\mathbf{W} = \begin{bmatrix} \sigma_{odo}^2 & 0 \\ 0 & \sigma_{gyro} \end{bmatrix}$$

$$\mathbf{G} = \frac{\partial f(\mathbf{x}, \mathbf{u}, \mathbf{w})}{\partial \mathbf{w}}\Big|_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}, \mathbf{u}=\mathbf{u}_k}$$

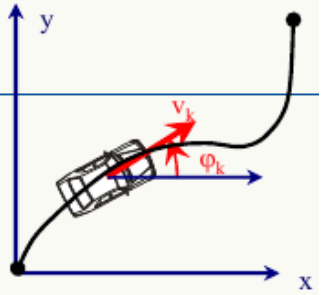
$$\mathbf{Q} = \mathbf{G}\mathbf{W}\mathbf{G}^T$$

KF EXAMPLE 3

$$\begin{aligned}\mathbf{x}_k^- &= f(\mathbf{x}_{k-1}, \mathbf{u}_k)\\ \mathbf{P}_k^- &= \mathbf{\Phi}\mathbf{P}_{k-1}\mathbf{\Phi}^T + \boxed{\mathbf{Q}}\end{aligned}$$

SYSTEM NOISE

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \end{bmatrix}$$



$$\mathbf{W} = \begin{bmatrix} \sigma_{odo}^2 & 0 \\ 0 & \sigma_{gyro} \end{bmatrix}$$

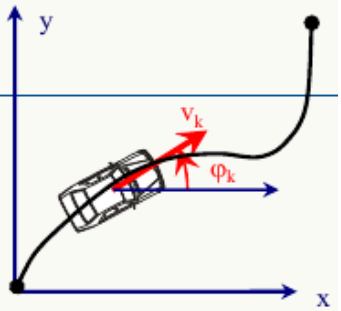
$$\mathbf{G} = \frac{\partial f(\mathbf{x}, \mathbf{u}, \mathbf{w})}{\partial \mathbf{w}} = \begin{bmatrix} \cos(\varphi_{k-1}) & 0 \\ \sin(\varphi_{k-1}) & 0 \\ 0 & \Delta t \end{bmatrix}_{\mathbf{x}=\hat{\mathbf{x}}_{k-1}, \mathbf{u}=\mathbf{u}_k}$$

$$\begin{aligned}\mathbf{Q} &= \mathbf{G}\mathbf{W}\mathbf{G}^T \\ &= \begin{bmatrix} \cos^2(\varphi_{k-1})\sigma_{odo}^2 & \cos(\varphi_{k-1})\sin(\varphi_{k-1})\sigma_{odo}^2 & 0 \\ \cos(\varphi_{k-1})\sin(\varphi_{k-1})\sigma_{odo}^2 & \sin^2(\varphi_{k-1})\sigma_{odo}^2 & 0 \\ 0 & 0 & \sigma_{gyro}^2\Delta t^2 \end{bmatrix}\end{aligned}$$

KF EXAMPLE 3

MEASUREMENT MODEL

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \varphi \end{bmatrix}$$



$$\begin{aligned} \mathbf{K}_k &= \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \boxed{\mathbf{R}})^{-1} \\ \mathbf{x}_k &= \mathbf{x}_k^- + \mathbf{K}_k(\mathbf{z}_k - \boxed{h(\mathbf{x}_k^-)}) \\ \mathbf{P}_k &= (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_k^- \end{aligned}$$

- Measurement is only GPS Position x_{gps}, y_{gps}

$$\mathbf{z}_k = \begin{bmatrix} x_{gps,k} \\ y_{gps,k} \end{bmatrix} = h(\mathbf{x}_k) = \mathbf{H}\mathbf{x}_k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

- Measurement Noise

$$\mathbf{R} = \begin{bmatrix} \sigma_{gps,x}^2 & 0 \\ 0 & \sigma_{gps,y}^2 \end{bmatrix}$$

Example 1 (**measuring** path increment, angular rate and gps , **modeling** constant angular rate and constant velocity)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \\ \dot{\varphi}_k \\ v_k \end{bmatrix} = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}} \Delta t \\ v_{k-1} + w_v \Delta t \end{bmatrix}$$

$$\begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ \Delta s_k \\ \omega_k \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

system noise

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\dot{\varphi}}^2 \Delta t^2 & 0 \\ 0 & 0 & 0 & 0 & \sigma_v^2 \Delta t^2 \end{bmatrix}$$

Example 2 (**measuring** acceleration, angular rate, and gps, **modelling** constant angular rate and constant acceleration)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \\ \dot{\varphi}_k \\ v_k \\ a_k \end{bmatrix} = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1}) v_{k-1} \Delta t \\ y_{k-1} + \sin(\varphi_{k-1}) v_{k-1} \Delta t \\ \varphi_{k-1} + \dot{\varphi}_{k-1} \Delta t \\ \dot{\varphi}_{k-1} + w_{\dot{\varphi}} \Delta t \\ v_{k-1} + a_{k-1} \Delta t \\ a_{k-1} + w_a \Delta t \end{bmatrix}$$

$$\begin{bmatrix} x_{gps,k} \\ y_{gps,k} \\ a_k \\ \omega_k \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \mathbf{x}_k$$

system noise

$$\mathbf{Q} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \sigma_{\dot{\varphi}}^2 \Delta t^2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & & 0 \sigma_a^2 \Delta t^2 \end{bmatrix}$$

Example 3 (**measuring** path increment, angular rate, and gps, using ang. rate and path inc. as **control input**)

$$\begin{bmatrix} x_k \\ y_k \\ \varphi_k \end{bmatrix} = \begin{bmatrix} x_{k-1} + \cos(\varphi_{k-1})(\Delta s_k + w_{odo}) \\ y_{k-1} + \sin(\varphi_{k-1})(\Delta s_k + w_{odo}) \\ \varphi_{k-1} + (\omega_{gyro,k} + w_{gyro}) \Delta t \end{bmatrix}$$

$$\begin{bmatrix} x_{gps,k} \\ y_{gps,k} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \mathbf{x}_k$$

system noise (from contol input)

$$\mathbf{Q} = \begin{bmatrix} \sin^2(\varphi_{k-1}) \sigma_{odo}^2 & \cos(\varphi_{k-1}) \sin(\varphi_{k-1}) \sigma_{odo}^2 & 0 \\ \cos(\varphi_{k-1}) \sin(\varphi_{k-1}) \sigma_{odo}^2 & \cos^2(\varphi_{k-1}) \sigma_{odo}^2 & 0 \\ 0 & 0 & \sigma_{gyro}^2 \Delta t^2 \end{bmatrix}$$